

$$M_{1,2}^{1,4} \left(M_{1,2,5,6}^{2,3,6,10} \cdot M_{1,2,3,4}^{5,7,8,9} - M_{1,2,4,6}^{2,3,6,10} \cdot M_{1,2,3,5}^{5,7,8,9} + M_{1,2,4,5}^{2,3,6,10} \cdot M_{1,2,3,6}^{5,7,8,9} \right. \\ \left. + M_{1,2,3,6}^{2,3,6,10} \cdot M_{1,2,4,5}^{5,7,8,9} - M_{1,2,3,5}^{2,3,6,10} \cdot M_{1,2,4,6}^{5,7,8,9} + M_{1,2,3,4}^{2,3,6,10} \cdot M_{1,2,5,6}^{5,7,8,9} \right).$$

Note that these are exactly the terms appearing in Example 3.9 (written in reverse order) and with the same signs as appear there.

Proof of Proposition 4.9. To compute the right-hand side of Equation (4.5), we first move $\nu - |\pi_i| = \nu - r - \nu_i$ vectors from the left side of the i th $\cap$ (that is, v_S) to its right side (that is, $v_{E \cup (\pi_i + n)}$). For each term τ of the right side and for each $i = 1, \dots, d-1$, let $S_{\tau,i} \subseteq S$ be the set of vectors that are *not* moved into the determinant and let $\overline{S_{\tau,i}} = S \setminus S_{\tau,i}$ be the set of vectors that are moved. This turns τ into an n by n determinant and picks up the sign $(-1)^{\text{inv}(\overline{S_{\tau,i}}, S_{\tau,i})}$ from the definition of $\cap$ in Equation (4.3). Next, we wedge together all of the vectors that are *not* moved together with the $v_{E \cup (\pi_d + n)}$ term. The result is a degree d monomial in Plücker coordinates together with a sign. The function (4.5) is the signed sum of these monomials.

Note that $|S_{\tau,i}| = \nu_i$ does not depend on τ . We define $S_{\tau,d}$ to be $S \setminus \bigcup_{i=1}^{d-1} S_{\tau,i}$. In order to get a nonzero evaluation, we need the disjoint union $S_{\tau,1} \sqcup \dots \sqcup S_{\tau,d}$ to equal S . In other words, we need the $S_{\tau,i}$ sets to be pairwise disjoint.

The term τ is then the product

$$(4.6) \quad \pm \prod_{i=1}^d \Delta_{\overline{S_{\tau,i}} \cup E \cup (\pi_i + n)}$$

of Plücker coordinates. Applying Φ^* to this, by Equation (4.2), we get

$$(4.7) \quad \pm \prod_{i=1}^d M_{[r] \cup S_{\tau,i}}^{\pi_i}.$$

Consider the jellyfish tableau $T(\tau) \in \mathcal{J}_r(\pi)$ that uses rows $S_{\tau,i} \cup [r]$ in column i . (Clearly, the correspondence between terms τ and jellyfish tableaux $T(\tau)$ is bijective.) The above monomial in matrix minors matches a the term in the definition of $[\pi_r]$ corresponding to $T(\tau)$. What remains to be checked is that the signs implicit in (4.5) match up with the signs from the definition of $[\pi]_r$ (i.e. the sign of the reading word of $T(\tau)$). For the purposes of this analysis, we say a quantity is *global* if it depends only on π , i.e. not on the specific jellyfish tableau $T(\tau)$ encoded by the specific sets $S_{\tau,1}, \dots, S_{\tau,d}$. For example, the ν_i 's are global quantities.

Now, fix τ and write $T = T(\tau)$. We first unpack the sign of the reading word of T . Let $\text{word}(T)$ be the reading word of T . We let $\widehat{T}$ be the “greedily top-justified left to right” jellyfish tableau in $\mathcal{J}_r(\pi)$. In Example 3.9, $\widehat{T}$ is the first tableau, and thus $\text{word}(\widehat{T}) = 2\ 5\ 1\ 3\ 7\ 4\ 6\ 10\ 8\ 9$. We define a permutation jp_T to be $\text{word}(\widehat{T})^{-1} \circ \text{word}(T)$; thus jp_T encodes the “jellyfish pattern” of T . In Example 3.9, the first 6 symbols of jp_T are always $1, \dots, 6$. (In general, the first rd symbols will be $1, \dots, rd$.) The last four symbols in our example are

$$7, 8, 9, 10 \text{ or } 7, 9, 8, 10 \text{ or } 7, 9, 10, 8 \text{ or } 9, 7, 8, 10 \text{ or } 9, 7, 10, 8 \text{ or } 9, 10, 7, 8,$$

respectively, in the same order as the jellyfish tableaux of Example 3.9.

Letting $S_i = S_{\tau,i}$ be the row indices $> r$ that are used column i of T as above, we see that

$$(4.8) \quad \text{inv}(\text{jp}_T) = \#\{u < v \in [n] : u \in S_i, v \in S_j \text{ and } i > j\},$$

which we can write as $\text{inv}(S_1, \dots, S_d)$ (with the number of inversions of a set-tuple defined to be the number of inversions of the permutation obtained by listing the elements in each set in increasing order). Since $\text{word}(\widehat{T})$ is a global quantity, it suffices to show that $\text{inv}(\text{jp}_T)$ differs from the exterior algebra sign we discuss below by a global sign.

Secondly, we unpack the exterior algebra signs associated with the calculation of the term τ in $[\pi]'_r$. The first type of sign contribution to τ comes due to bringing the set S_i to the back of the ordered set S ; hence, this first sign is

$$\prod_{i=1}^{d-1} (-1)^{\text{inv}(\overline{S_i}, S_i)}.$$

The second sign is the sign $(-1)^{\text{inv}(S_1, \dots, S_{d-1})}$ coming from wedging together the unmoved vectors for $i = 1, \dots, d-1$ in the wedge product.

We have $\text{inv}(S_1, \dots, S_d) = \text{inv}(S_1, \dots, S_{d-1}) + \text{inv}(\overline{S_d}, S_d)$. Therefore, the sign $(-1)^{\text{inv}(S_1, \dots, S_d)}$ computed from the jellyfish tableau formula differs from the sign computed from Equation (4.3) by

$$(-1)^{\sum_{i=1}^d \text{inv}(\overline{S_i}, S_i)}.$$

Hence, it suffices to show that this number is a global quantity. A pair u, v with $u \in S_i$ and $v \in S_j$ appears exactly once in this sum (it appears in the term indexed by i if $v < u$ and otherwise appears in the term indexed by j). The number of such pairs only depends on the block sizes and thus is a global quantity. (Specifically, it is $\frac{1}{2} \sum_{i=1}^d \nu_i(\nu - |\pi_i|)$.) This completes the proof.

Note that we have in fact shown that the sign appearing in the proposition is

$$(-1)^{\text{inv}(\text{word}(\widehat{T})) + \frac{1}{2} \sum_{i=1}^d \nu_i(\nu - |\pi_i|)}. \quad \square$$

4.5. JELLYFISH INVARIANTS AS TENSOR DIAGRAM INVARIANTS. As explained in Remark 4.6, any Grassmann–Cayley expression obtained by iterated composition of $\wedge$ and $\cap$ operations can be expressed as a tensor diagram invariant. In this section, we give a combinatorial construction of the tensor diagram that underlies the expression (4.5).

DEFINITION 4.11. *Given an ordered set partition $\pi \in \mathcal{OP}(n, d, r)$, we associate a tensor diagram $W_{\pi, r}$ of type $(n, 2n)$ using the following steps. Recall we have set $S = [r+1, \nu]$ and $E = [\nu+1, n]$.*

- (1) *For $i \in [d]$, introduce a interior white vertex w_i and draw edges from w_i to every boundary vertex in E , as well as to every boundary vertex in $\pi_i + n$. All of these edges receive weight 1.*
- (2) *For $i \in [d-1]$, introduce an interior white vertex u_i and draw edges from u_i to every boundary vertex in S . These edges also all receive weight 1.*
- (3) *For $i \in [d-1]$, introduce an interior black vertex b_i . For $i \in [d-1]$, draw an edge between each b_i and the corresponding w_i and give this edge weight $\nu - |\pi_i|$. Also, draw an edge between each b_i and the corresponding u_i , giving this edge weight $n - \nu + r = rd$. Finally, draw an edge between each b_i and the vertex w_d , giving this edge weight ν_i .*

See Figure 4 for a schematic diagram of this definition and Figure 5 for an example. Clearly the tensor diagram $W_{\pi, r}$ is typically highly nonplanar.

LEMMA 4.12. *The diagram $W_{\pi, r}$ constructed in Definition 4.11 is a tensor diagram of type $(n, 2n)$.*

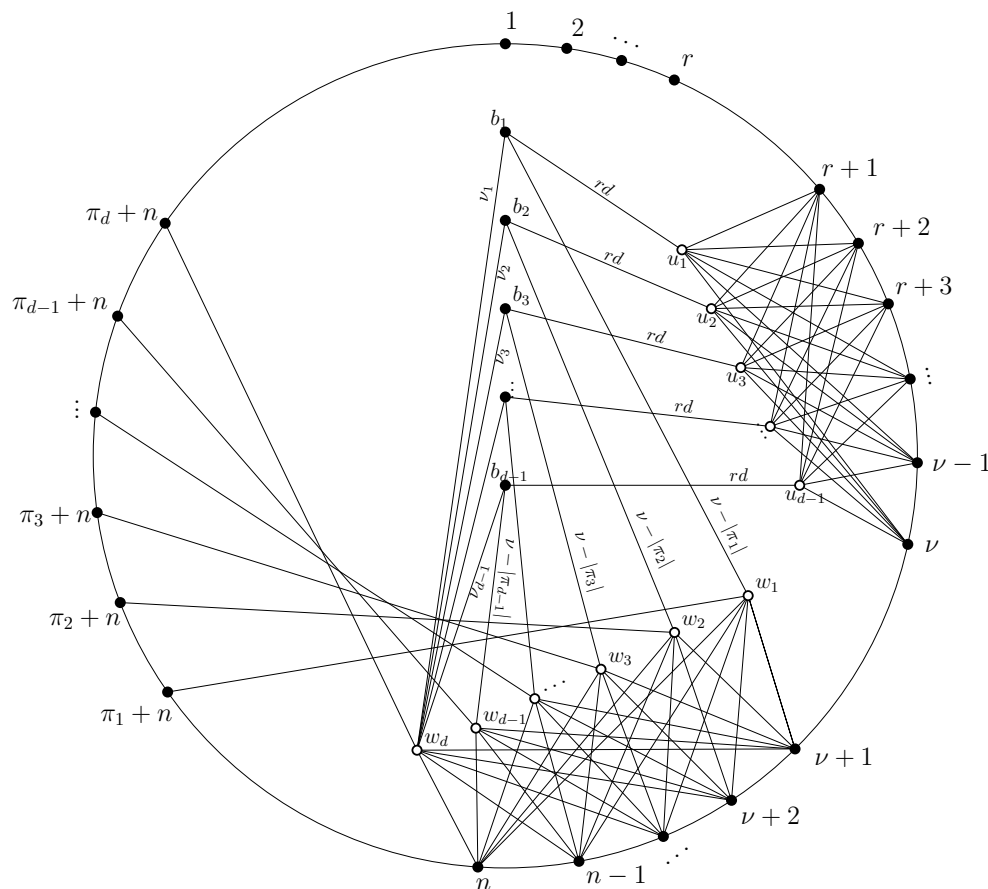


FIGURE 4. A schematic tensor diagram $W_{\pi,r}$ illustrating the algorithm in Definition 4.11. Vertices $1, \dots, r$ are not used. Vertices in $S = \{r + 1, \dots, \nu\}$ and $E = \{\nu + 1, \dots, n\}$ are bookkeeping vertices such that each u_i connects to all vertices in S and each w_i connects to all vertices in E . The vertices $\pi_i + n$ are schematic, and they encode the set partition π in the sense that w_i is connected to all of the $|\pi_i|$ boundary vertices in $\pi_i + n$. Note these vertices will not usually be cyclically ordered as we have drawn them here. See Figure 5 for an explicit example.

Proof. It is enough to verify that sum of weights around every internal vertex is n .
For $i \in [d - 1]$, each w_i has

- $|\pi_i|$ edges of weight 1 connecting w_i to the boundary vertices in π_i ,
- $n - \nu$ edges of weight 1 connecting w_i to the boundary vertices in E , and
- 1 edge of weight $\nu - |\pi_i|$ connecting w_i to b_i .

Thus, the edges incident at w_i have weights summing to

$$n - \nu + \nu - |\pi_i| + |\pi_i| = n,$$

as desired.

The vertex w_d has

- 1 edge of weight ν_i for each $i \in [d - 1]$ connecting w_d to b_i ,

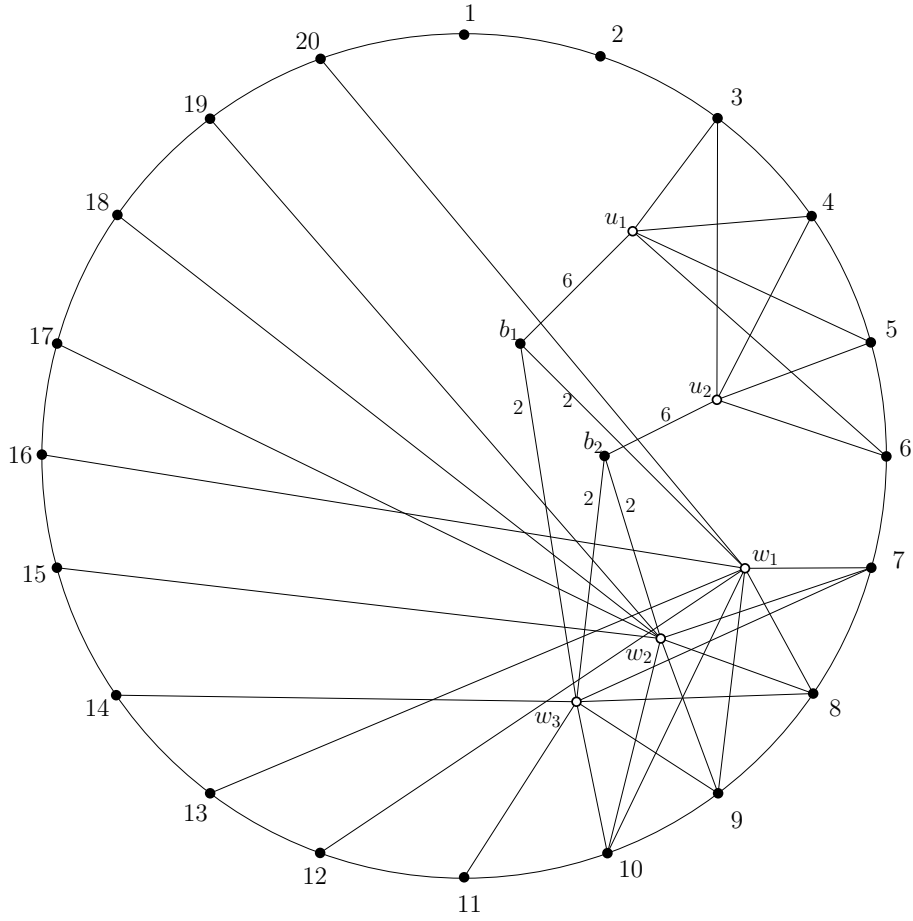


FIGURE 5. The tensor diagram obtained by Definition 4.11 from ordered set partition $\pi = (2\ 3\ 6\ 10\ |\ 5\ 7\ 8\ 9\ |\ 1\ 4)$ from Examples 3.9 and 4.10. Note for example that w_1 connects to boundary vertices 12, 13, 16, and 20, which is $\pi_1 + n$ in this example.

- $|\pi_d| = \nu_d + r$ edges of weight 1 connecting w_d to the boundary vertices in π_d , and
- $n - \nu = rd - r$ edges of weight 1 connecting w_i to the boundary vertices in E .

Thus, the edges incident at w_d have weights summing to

$$\left(\sum_{i=1}^{d-1} \nu_i \right) + \nu_d + r + rd - r = rd + \sum_{i=1}^d \nu_i = n,$$

as desired.

Each vertex u_i has

- 1 edge of weight $n - \nu + r$ connecting u_i to the corresponding b_i and
- $\nu - r$ edges of weight 1 connecting u_i to the boundary vertices in S ,

giving a sum of n .

Finally, each vertex b_i has

- 1 edge of weight $n - \nu + r$ connecting b_i and u_i ,
- 1 edge of weight $\nu - |\pi_i| = \nu - (r + \nu_i)$ connecting b_i and w_i , and
- 1 edge of weight ν_i connecting b_i and w_d .

Thus, the edges incident to b_i have weight summing to

$$n - \nu + r + \nu - (r + \nu_i) + \nu_i = n,$$

completing the proof. $\square$

PROPOSITION 4.13. *The tensor diagram $W_{\pi,r}$ encodes the Grassmann–Cayley expression (4.5) up to sign.*

Proof. The structure of the proof is as follows. First, we will interpret the Grassmann–Cayley expression (4.3) as a sequence of $\wedge$ and dual exterior product operations, giving rise to a tagged tensor diagram W_1 . By construction, this tagged tensor diagram W_1 computes the Grassmann–Cayley expression. Second, we will choose a tagging W_2 of $W_{\pi,r}$ by choosing a perfect orientation of the latter graph. Then we can apply [6, Lemma 5.4], which asserts that the tagged tensor invariant $[W_2]$ agrees with $[W_{\pi,r}]$ up to sign. Finally we will argue that $[W_1]$ and $[W_2]$ agree up to sign, completing the proof.

For the first step, we express the Grassmann–Cayley expression in terms of tagged tensor diagram constructions. The tensor $v_{E \cup (\pi_i + n)}$ is diagrammatically encoded by an interior white vertex w_i with incoming edges of weight 1 from the boundary vertices in $E \cup (\pi_i + n)$, for $i = 1, \dots, d$ and with one outgoing edge of weight $|E| + |\pi_i|$. The tensor v_S is encoded by a white vertex u_i with incoming edges of weight 1 from the boundary vertices in S and with one outgoing edge of weight $|S|$, for $i = 1, \dots, d-1$. The cap $v_S \cap v_{E \cup (\pi_i + n)}$ of these tensors is encoded by a black vertex b_i whose unique incoming edge emanates from vertex u_i and which has two outgoing edges. The first of these outgoing edges has weight $n - |E| - |\pi_i| = \nu - |\pi_i|$ and meets the outgoing edge emanating from w_i at a tag; this tag represents the explicit determinant in the definition of the meet operation (4.3). The second outgoing edge emanating from b_i has weight ν_i , representing the wedge after the determinant in (4.3). It meets the edge emanating from w_d at a tag, computing the wedge of the tensors just mentioned with the last factor $v_{E \cup (\pi_d + n)}$ in (4.5). Call the resulting tagged tensor diagram W_1 .

Second, we obtain a perfect orientation of $W_{\pi,r}$ as follows. Identify the sets S and E with their corresponding boundary vertices. We orient edges $S \rightarrow u_i$ for $i = 1, \dots, d-1$; $u_i \rightarrow b_i$ for $i = 1, \dots, d-1$; $b_i \rightarrow w_i$ and $b_i \rightarrow w_d$ for $i = 1, \dots, d-1$. We orient edges $E \setminus \{n\} \rightarrow w_i$ and $w_i \rightarrow n$ for $i = 1, \dots, d$. Applying the construction from Section 4.3, we obtain a tagged diagram W_2 .

To complete the proof, note that W_1 and W_2 differ only by applying tag migration moves [1, Equation 2.7], migrating the tagged edge between boundary vertex n and w_i to the edges between the b_i 's and w_i 's. Tag migration moves only change sign, completing the proof. $\square$

We now conclude the main result of this section. That is, up to sign and application of the translation map Φ^* , r -jellyfish invariants associated to ordered set partitions are tensor diagram invariants in the classical sense.

THEOREM 4.14. *For any ordered set partition $\pi \in \mathcal{OP}(n, d, r)$, the corresponding r -jellyfish invariant is a tensor invariant; precisely, we have*

$$[\pi]_r = \pm \Phi^*([W_{\pi,r}]).$$

Proof. Combining Propositions 4.9 and 4.13, we obtain the desired result. $\square$

4.6. JELLYFISH INVARIANTS LIE IN FLAMINGO SPECHT MODULES. We now use the commutativity of the $\cap$ operation to establish an important property of r -jellyfish invariants. The $r = 2$ case is [21, Lemma 3.14].

DEFINITION 5.2. Let $T \in \overline{\mathcal{J}}_r(\pi)$. Define the **inversion number** $\text{inv}(T)$ as the number of inversions in the row reading word (left to right, top to bottom), except we do not count inversions within columns. Define $\text{sgn}(T) = (-1)^{\text{inv}(T)}$.

Note that $\mathcal{J}_r(\pi) \subsetneq \overline{\mathcal{J}}_r(\pi)$ and Definition 3.2 agrees with the restriction of Definition 5.2, since there are no inversions within columns of a tableau in $\mathcal{J}_r(\pi)$.

LEMMA 5.3. Suppose $U \in \mathcal{J}_r(\pi)$ and $T \in \overline{\mathcal{J}}_r(U)$. Then $\text{sgn}(U) = \text{sgn}(T)$.

Proof. This argument is the same as for the analogous statement in [21, Lemma 3.9], and we will reproduce it here. Suppose $T \in \overline{\mathcal{J}}(U)$ and suppose T' is obtained from T by permuting the labels within columns. We show $\text{sgn}(T) = \text{sgn}(T')$ (and hence, by repeated application, $\text{sgn}(T) = \text{sgn}(U)$). Note $\text{inv}(T)$ may not equal $\text{inv}(T')$, but they have the same parity.

It is enough to prove the result in the case that T' is obtained from T by interchanging two entries i and j in a single column. Recall that $\text{inv}(T)$ is the number of inversions in the row reading word of T , except ignoring inversions within columns. For each pair of entries k, ℓ of T , define

$$\iota(k, \ell) = \begin{cases} -1, & \text{if } k \text{ and } \ell \text{ are inverted in the reading word of } T; \\ 1, & \text{otherwise.} \end{cases}$$

Consider any entry k outside of the column containing i and j . Interchanging the elements i and j in T swaps their positions in the reading word. Therefore, the sum $\iota(i, k) + \iota(j, k) = 0$. Hence, this swap does not affect the parity of the number of inversions involving (i, k) and (j, k) . It does change whether (i, j) is an inversion in the reading word; however, since they are in the same column, this does not affect the inversion number of the jellyfish tableau. Hence, $\text{sgn}(T) = \text{sgn}(T')$. $\square$

COROLLARY 5.4. Suppose that $\pi, \pi' \in \mathcal{OP}(n, d, r)$ are ordered set partitions with the same underlying unordered set partition, so that there exists $\sigma \in \mathfrak{S}_d$ with $\sigma(\pi) = \pi'$. Then,

$$[\pi]_r = \text{sgn}(\sigma)^r [\pi']_r.$$

Proof. This is immediate from combining Lemmas 3.3 and 5.3. $\square$

5.2. A $(2^r + 1)$ -TERM RECURRENCE. In this subsection, we state and prove a useful recurrence, Theorem 5.5, which we apply in the next subsection to give an alternate proof of Theorem 4.15, that the r -jellyfish invariants $[\pi]_r$ are contained in the appropriate Specht modules. The $r = 2$ case of this recurrence appeared as [21, Theorem 3.11] with a direct proof by Laplace expansion of determinants. Here, we identify the recurrence as an avatar of the classical Plücker relation. The description of signs appearing in Theorem 5.5 is very different from the classical Plücker sign calculation, so this perspective may be generally useful as a new way to think about the signs appearing in Plücker relations.

THEOREM 5.5. Partition $\{1, \dots, n\}$ into three nonempty sets: A, B, C , where $|C| = r$. Then

$$(5.1) \quad [(A \cup B \mid C)]_r = \sum_{S \subseteq C} (-1)^{|S|} [(A \cup S \mid B \cup (C \setminus S))]_r$$

To illustrate Theorem 5.5, let $r = 3$ and suppose $C = \{c_1, c_2, c_3\}$. Figure 6 gives a diagrammatic representation of the 9-term recurrence:

$$\begin{aligned}
 & [(A \cup B \mid C)]_3 = \\
 & [(A \mid B \cup C)]_3 - [(A \cup \{c_1\} \mid B \cup \{c_2, c_3\})]_3 - [(A \cup \{c_2\} \mid B \cup \{c_1, c_3\})]_3 \\
 & - [(A \cup \{c_3\} \mid B \cup \{c_1, c_2\})]_3 + [(A \cup \{c_1, c_2\} \mid B \cup \{c_3\})]_3 + [(A \cup \{c_1, c_3\} \mid B \cup \{c_2\})]_3 \\
 & + [(A \cup \{c_2, c_3\} \mid B \cup \{c_1\})]_3 - [(A \cup C \mid B)]_3
 \end{aligned}$$

FIGURE 6. An illustration of the recurrence of Theorem 5.5 in the case $r = 3$.

Proof of Theorem 5.5. We prove Theorem 5.5 using Plücker coordinates and relations. As a shorthand, for W any set of positive integers, we write $W' = \{w + n : w \in W\}$.

First, we rewrite the left side of (5.1) in terms of Plücker variables, using Equation (4.2). By definition, there is only one jellyfish tableau for the ordered set partition $(A \cup B \mid C)$, since $|C| = r$. Call this tableau $\hat{T}$. We have

$$\begin{aligned}
 (5.2) \quad & [(A \cup B \mid C)]_r = \sum_{T \in \mathcal{J}_r(A \cup B \mid C)} \text{sgn}(T) J(T) \\
 & = \text{sgn}(\hat{T}) M_{R_{A \cup B}(\hat{T})}^{A \cup B} \cdot M_{R_C(\hat{T})}^C \\
 & = \text{sgn}(\hat{T}) (-1)^{|A \cup B|} \Delta_{([n] \setminus R_{A \cup B}(\hat{T}))A' B'}(\Phi(M)) \cdot (-1)^{|C|} \Delta_{([n] \setminus R_C(\hat{T}))C'}(\Phi(M)) \\
 & = (-1)^n \text{sgn}(\hat{T}) \Delta_{[n-r+1, n]A' B'}(\Phi(M)) \cdot \Delta_{[r+1, n]C'}(\Phi(M))
 \end{aligned}$$

Here, the first and second equalities are by definition, while the third is using (4.2) and the fourth line collects the sign.

We will apply the classical Plücker relation in a form that we borrow from [8, Eqn. 9.1]:

$$(5.3) \quad \Delta_{i_1, i_2, \dots, i_d} \cdot \Delta_{j_1, j_2, \dots, j_d} = \sum \Delta_{i'_1, i'_2, \dots, i'_d} \cdot \Delta_{j'_1, j'_2, \dots, j'_d},$$

where the sum is over all pairs obtained by interchanging a fixed set of k of the subscripts $j_1, j_2, \dots, j_d$ with k of the subscripts in $i_1, i_2, \dots, i_d$, maintaining the order in each. Note that whenever the resulting subscripts are rearranged, signs are introduced. In our setting, we will have $k = |B|$.

Before applying the Plücker relation, we rearrange the last line of (5.2) in order to better match (5.3), obtaining:

(5.4)

$$\begin{aligned} [(A \cup B \mid C)]_r &= (-1)^n \operatorname{sgn}(\widehat{T}) \Delta_{[r+1,n]C'}(\Phi(M)) \cdot \Delta_{[n-r+1,n]A'B'}(\Phi(M)) \\ &= (-1)^n \operatorname{sgn}(\widehat{T}) \Delta_{[n-r+1,n]C'[r+1,n-r]}(\Phi(M)) \cdot \Delta_{[n-r+1,n]A'B'}(\Phi(M)). \end{aligned}$$

In the second equality above, we also moved the interval $[r+1, n-r]$ to the end of its Plücker coordinate to simplify things later. This move does not affect the global sign because each element in $[r+1, n-r]$ moves past the r elements of $[n-r+1, n]$ and the r elements of C' , for an even total number of swaps.

The particular Plücker relation we apply is the one in which B' is the fixed set of k subscripts that gets interchanged. What happens is that the first ℓ indices in B' interchange with a subset Z of $[r+1, n]$ and the rest interchange with a subset S' of C' . We can match each term from the right side of (5.3) with the column-permuted jellyfish tableau that is obtained from $\widehat{T}$ by moving the elements of B from column 1 to column 2, moving the elements of S from column 2 to column 1, and so that the elements of Z are the row indices of the empty boxes in column 1. By the definition of jellyfish tableaux, this also determines the row indices of the empty boxes in column 2. See Figure 7 for an example of this construction (the unexplained notations in that figure will be introduced later for a sign calculation). By Lemma 5.3, the ordering of the labels within each column is immaterial; however, the ordering we impose here will be useful later. Thus, each choice of subsets S and Z , corresponds to a unique jellyfish tableau contributing to the right hand side of (5.1) by sorting the columns in increasing order. Conversely, one can check that each of the jellyfish tableaux corresponds to a term of the Plücker relation.

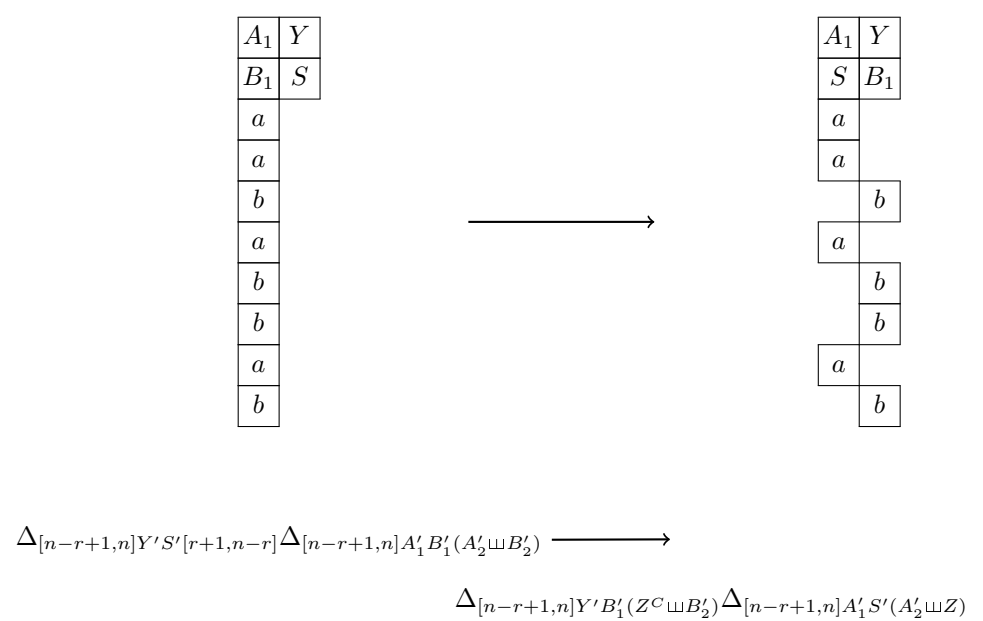


FIGURE 7. An illustration of the proof of Theorem 5.5.

Theorem 5.5 is then proved, except for comparing the signs from the jellyfish tableaux with the signs arising from sorting indices in (5.3).

As a warm up, we first consider the case $|A \cup B| = r$. Note that in this case, $n = 2r$, so $[r + 1, n - r] = \emptyset$. We can assume our Plücker monomial from (5.4) looks like $\Delta_{[n-r+1, n]Y'S'}\Delta_{[n-r+1, n]A'B'}$, where $Y = C \setminus S$. Since the elements of the subscripts on these Plücker variables are out of order, there is an associated sign: -1 to the number of inversions. We want to see how this sign changes when we rearrange indices to obtain $\Delta_{[n-r+1, n]Y'B'}\Delta_{[n-r+1, n]A'S'}$. Note that for each choice of S , we are choosing a different ordering in the initial Plücker monomial from (5.4), but that this is acceptable, since we only need to understand the relation between the signs of the two monomials of interest. In comparing the signs of these monomials, it is straightforward to see that the signs differ by -1 to the number of pairs $(t_1, t_2) \in (B \cup S) \times (Y \cup A)$ with $t_1 < t_2$.

We now perform the corresponding sign calculation for the corresponding jellyfish tableaux. By Lemma 5.3, we can instead compare the column-permuted jellyfish tableaux

$$\begin{array}{|c|c|} \hline A & Y \\ \hline B & S \\ \hline \end{array} \text{ and } \begin{array}{|c|c|} \hline A & Y \\ \hline S & B \\ \hline \end{array},$$

where the elements of each set are arranged vertically in a column in some fixed order. Since inversions in a column do not contribute to the sign of a jellyfish tableau, here there is a sign change for each pair $(t_1, t_2) \in (B \cup S) \times (Y \cup A)$ with $t_1 < t_2$, matching the sign changes from shuffling indices in the Plücker monomials, as described in the previous paragraph; in addition, there is a sign change for each of the bottom $|B| = |S|$ rows, as desired to match the sign appearing on the right-hand side of (5.1). This completes the proof in the special case $|A \cup B| = r$.

Now we consider the general case of this sign calculation. Let B_1 be the first $|S|$ elements of B and let $B_2 = B \setminus B_1$. Similarly, let A_1 be the first $r - |S|$ elements of A and let $A_2 = A \setminus A_1$.

We study the column-permuted version of $\hat{T}$ for which the left column has A_1 (in increasing order) above B_1 (in increasing order) occupying the body of the jellyfish, and then A_2 and B_2 in the following rows, shuffled together so that B_2 occupies the rows indexed by Z . Write $A_2 \sqcup B_2$ for this ordering of $A_2 \cup B_2$. We also write Z^C for $[r + 1, n - r] \setminus Z$. The right column of our column-permuted $\hat{T}$ has $Y = C \setminus S$ (in increasing order) above S (in increasing order). See the left tableau of Figure 7 for an illustration.

We can assume our corresponding Plücker monomial from (5.4) is written in a similar way, as

$$\Delta_{[n-r+1, n]Y'S'}\Delta_{[r+1, n-r]A_1'B_1'(A_2' \sqcup B_2')},$$

where $Y = C \setminus S$. Then we want to compare the sign of this with the sign for

$$\Delta_{[n-r+1, n]Y'B_1'(Z^C \sqcup B_2')}\Delta_{[n-r+1, n]A_1'S'(A_2' \sqcup Z)}.$$

Again, note that for each choice of S , we are choosing a different ordering in the corresponding Plücker monomial. In comparing the signs of these monomials, it is again straightforward to see that swapping B_1' with S' results in a sign difference of -1 to the number of pairs $(t_1, t_2) \in (B_1 \cup S) \times (Y \cup A_1)$ with $t_1 < t_2$ together with -1 to the number of pairs $(t_1, t_2) \in (B_1 \cup S) \times A_2$ with $t_1 > t_2$. Furthermore, the sign difference in swapping B_2' and Z comes from the fact that all the elements of B_2' are larger than the elements of Z^C and all the elements of Z are smaller than the elements of A_2' . So after the swap, all the elements of B_2' have inversions with the elements of $[r + 1, n - r]$ to their right, while all the elements of Z have inversions with the elements of A_2' to their left. Since the positions occupied by B_2' and Z are corresponding, this results in a total of $n - 2r$ inversions. Since $2r$ is even, this results in a contribution of $(-1)^n$.

We now compute the corresponding sign difference between the corresponding column-permuted jellyfish tableaux. See Figure 7 for a visual comparison of these

tableaux. It is again straightforward to see that swapping B_1 with S results in a sign difference of -1 to the number of pairs $(t_1, t_2) \in (B_1 \cup S) \times (Y \cup A_1)$ with $t_1 < t_2$ together with -1 to the number of pairs $(t_1, t_2) \in (B_1 \cup S) \times A_2$ with $t_1 > t_2$. Furthermore, there is a sign change for each of the bottom $|B_1| = |S|$ rows.

So, continuing the calculation from (5.4), we have that

$$\begin{aligned} [(A \cup B \mid C)]_r &= (-1)^n \operatorname{sgn}(\widehat{T}) \sum_{S \subseteq C} \operatorname{sgn}(\widehat{T}) (-1)^n (-1)^{|S|} [(A \cup S \mid B \cup (C \setminus S))]_r \\ &= \sum_{S \subseteq C} (-1)^{|S|} [(A \cup S \mid B \cup (C \setminus S))]_r \end{aligned}$$

as desired. $\square$

We now note that we can use this rule in set partitions with more than two blocks by fixing all blocks except those involving A , B , and C .

COROLLARY 5.6. *Partition $\{1, \dots, n\}$ into $d+1$ nonempty sets: $\pi_1, \pi_2, \dots, \pi_{d-2}, A, B$, and C , where $|C| = r$. Then*

$$(5.5) \quad [(\pi_1 \mid \dots \mid \pi_{d-2} \mid A \cup B \mid C)]_r = \sum_{S \subseteq C} (-1)^{|S|} [(\pi_1 \mid \dots \mid \pi_{d-2} \mid A \cup S \mid B \cup (C \setminus S))]_r.$$

Proof. The arguments in the proof of Theorem 5.5 apply, since the Plücker coordinates corresponding to $\pi_1, \pi_2, \dots, \pi_{d-2}$ factor out. $\square$

5.3. r -JELLYFISH INVARIANTS FOR FLAMINGO SPECHT MODULES. In this subsection, we give an alternate proof of Theorem 4.15 using the recurrence of Theorem 5.5. We also describe the action of the symmetric group $\mathfrak{S}_n$. Finally, we show linear independence of r -jellyfish invariants for $r > 1$.

Alternate proof of Theorem 4.15. Recall from Section 2.3 that $S^{(d^r, 1^{n-rd})}$ is generated by fillings of partition shape $(d^r, 1^{n-rd})$ where each of $\{1, \dots, n\}$ is used exactly once, and the invariant is given by multiplying the Plücker variables corresponding to the columns of the filling. Note that if $\pi \in \mathcal{OP}(n, d, r)$ has $d-1$ blocks of size r , then $[\pi]_r$ is a single term where the corresponding jellyfish tableau is a filling of partition shape $(d^r, 1^{n-rd})$ with each of $\{1, \dots, n\}$ used exactly once. Thus clearly $[\pi]_r \in S^{(d^r, 1^{n-rd})}$ by the explicit description of Specht modules from Section 2.3. We use this fact repeatedly in our argument below.

We first consider the case where $d = 2$. That is, we show that the invariant $[\pi]_r$ of any ordered set partition π of n with 2 blocks and parts of size at least r is in $S^{(2^r, 1^{n-2r})}$. We show this by induction on the number of elements in the smaller block.

The first non-trivial case is when the smaller block has $r+1$ elements. Suppose that $\pi = (A \mid T)$ with $T = \{t_1, t_2, \dots, t_{r+1}\}$ and $|A| \geq r+1$. We apply Equation (5.1) with A as given, $B = \{t_1\}$, and $C = T \setminus \{t_1\}$. Then Equation (5.1) yields

$$\begin{aligned} [(A \cup \{t_1\} \mid \{t_2, \dots, t_{r+1}\})]_r &= \sum_{S \subseteq T \setminus \{t_1\}} (-1)^{|S|} [(A \cup S \mid \{t_1\} \cup (T \setminus \{t_1\}) \setminus S)]_r \\ &= [(A \mid T)]_r - \sum_{i \in \{2, \dots, r+1\}} [(A \cup \{t_i\} \mid T \setminus \{t_i\})]_r \\ &\quad + \sum_{\substack{S \subseteq T \setminus \{t_1\} \\ |S| > 1}} (-1)^{|S|} [(A \cup S \mid T \setminus S)]_r. \end{aligned}$$

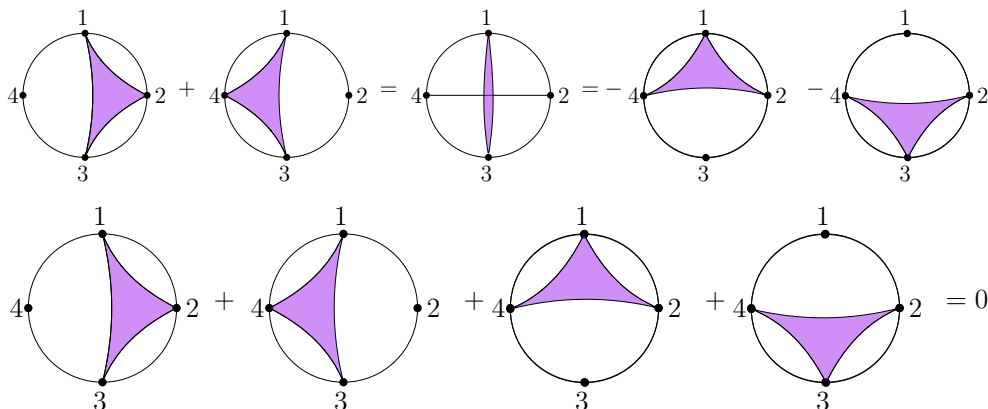
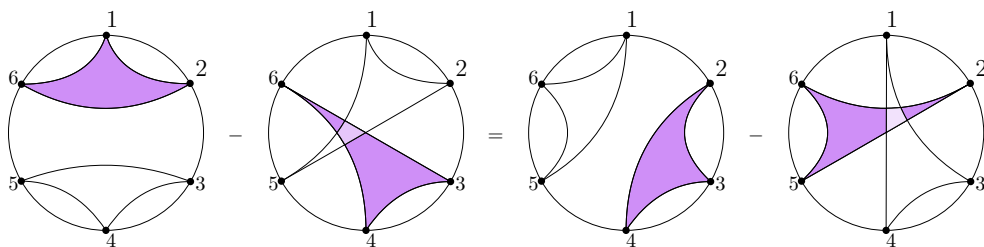


FIGURE 8. The top image shows how to resolve a crossing in two different ways when $r = 1$. The second image shows a linear dependence among noncrossing partitions when $r = 1$. In both images, the block colored purple is the first block of the ordered set partition.



Towards constructing web bases for general flamingo Specht modules, we conjecture the following extension of Theorem 5.9. For $r = 3$, the set in question is the set of noncrossing partitions, and hence follows from Theorem 5.9. We have verified the conjecture for some higher r by computer calculation.

CONJECTURE 5.13. *Let $r \geq 3$ and let $\mathcal{S} \subset \Pi(n, d, r)$ denote the collection of set partitions that can be made noncrossing by the application of at most $r - 3$ adjacent transpositions. For each set partition $\gamma \in \mathcal{S}$, order the blocks in any way to create a corresponding ordered set partition π_γ . Then the set $\{[\pi_\gamma]_r : \gamma \in \mathcal{S}\}$ is linearly independent.*

We suspect that the recurrence of Theorem 5.5 will be useful in approaching Conjecture 5.13, as it allows one to rewrite the polynomial for an ordered set partition by a linear combination of polynomials for ordered set partitions that are closer to noncrossing.⁽¹⁾

It would be convenient if for $r > 1$ and $\pi \in \mathcal{OP}(n, d, r)$, the set of r -jellyfish polynomials of rotations of π were linearly independent. However, this is not true. For example, consider $\pi = (1\ 2\ 3\ 5 \mid 4\ 6)$ and $r = 2$. Then the rotation orbit of π consists of 6 distinct set partitions. However, one may check that the corresponding 2-jellyfish invariants span only a 5-dimensional space.

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⁽¹⁾After this preprint was posted, Jesse Kim built upon the ideas of this manuscript and found a basis for flamingo Specht modules in terms of jellyfish invariants of r -weakly noncrossing set partitions [15].

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REFERENCES

- [1] Sabin Cautis, Joel Kamnitzer, and Scott Morrison, *Webs and quantum skew Howe duality*, Math. Ann. **360** (2014), no. 1-2, 351–390.
- [2] Ben Elias, *Light ladders and clasp conjectures*, 2015, <https://arxiv.org/abs/1510.06840>.
- [3] Sergey Fomin and Pavlo Pylyavskyy, *Tensor diagrams and cluster algebras*, Adv. Math. **300** (2016), 717–787.
- [4] Bruce Fontaine, *Generating basis webs for SL_n* , Adv. Math. **229** (2012), no. 5, 2792–2817.
- [5] Chris Fraser, *Webs and canonical bases in degree two*, Comb. Theory **3** (2023), no. 3, article no. 11 (26 pages).
- [6] Chris Fraser, Thomas Lam, and Ian Le, *From dimers to webs*, Trans. Amer. Math. Soc. **371** (2019), no. 9, 6087–6124.
- [7] Chris Fraser and Pavlo Pylyavskyy, *Tensor diagrams and cluster combinatorics at punctures*, Adv. Math. **412** (2023), article no. 108796 (83 pages).
- [8] William Fulton, *Young tableaux. With applications to representation theory and geometry*, London Mathematical Society Student Texts, vol. 35, Cambridge University Press, Cambridge, 1997.
- [9] William Fulton and Joe Harris, *Representation theory. A first course*, Readings in Mathematics, Graduate Texts in Mathematics, vol. 129, Springer-Verlag, New York, 1991.
- [10] Christian Gaetz, Oliver Pechenik, Stephan Pfannerer, Jessica Striker, and Joshua P. Swanson, *Rotation-invariant web bases from hourglass plabic graphs*, 2023, <https://arxiv.org/abs/2306.12501>.
- [11] Christian Gaetz, Oliver Pechenik, Stephan Pfannerer, Jessica Striker, and Joshua P. Swanson, *Web bases in degree two from hourglass plabic graphs*, 2024, <https://arxiv.org/abs/2402.13978>.
- [12] Byung-Hak Hwang, Jihyeug Jang, and Jaeseong Oh, *A combinatorial model for the transition matrix between the Specht and SL_2 -web bases*, Forum Math. Sigma **11** (2023), article no. e82 (17 pages).
- [13] Mee Seong Im and Jieru Zhu, *Transitioning Between Tableaux and Spider Bases for Specht Modules*, Algebr. Represent. Theory **25** (2022), no. 2, 387–399.
- [14] Jesse Kim, *An embedding of the skein action on set partitions into the skein action on matchings*, Electron. J. Combin. **31** (2024), no. 1, article no. 1.17 (17 pages).
- [15] Jesse Kim, *Rotation invariant webs for three row flamingo Specht modules*, 2024, <https://arxiv.org/abs/2402.11994>.
- [16] Jesse Kim and Brendon Rhoades, *Set Partitions, Fermions, and Skein Relations*, Int. Math. Res. Not. IMRN **2023** (2023), no. 11, 9427–9480.
- [17] Joseph P. S. Kung and Gian-Carlo Rota, *The invariant theory of binary forms*, Bull. Amer. Math. Soc. (N.S.) **10** (1984), no. 1, 27–85.
- [18] Greg Kuperberg, *Spiders for rank 2 Lie algebras*, Comm. Math. Phys. **180** (1996), no. 1, 109–151.
- [19] G. Lusztig, *Canonical bases arising from quantized enveloping algebras*, J. Amer. Math. Soc. **3** (1990), no. 2, 447–498.
- [20] Rebecca Patrias and Oliver Pechenik, *Tableau evacuation and webs*, Proc. Amer. Math. Soc. Ser. B **10** (2023), 341–352.
- [21] Rebecca Patrias, Oliver Pechenik, and Jessica Striker, *A web basis of invariant polynomials from noncrossing partitions*, Adv. Math. **408** (2022), article no. 108603 (33 pages).
- [22] Oliver Pechenik, *Cyclic sieving of increasing tableaux and small Schröder paths*, J. Combin. Theory Ser. A **125** (2014), 357–378.
- [23] T. Kyle Petersen, Pavlo Pylyavskyy, and Brendon Rhoades, *Promotion and cyclic sieving via webs*, J. Algebraic Combin. **30** (2009), no. 1, 19–41.
- [24] Kevin Purbhoo and Shelley Wu, *Folding rotationally symmetric tableaux via webs*, Ann. Comb. **28** (2024), no. 1, 93–119.
- [25] Brendon Rhoades, *Cyclic sieving, promotion, and representation theory*, J. Combin. Theory Ser. A **117** (2010), no. 1, 38–76.
- [26] Brendon Rhoades, *A skein action of the symmetric group on noncrossing partitions*, J. Algebraic Combin. **45** (2017), no. 1, 81–127.

- [27] Brendon Rhoades, *The polytabloid basis expands positively into the web basis*, Forum Math. Sigma **7** (2019), article no. e26 (8 pages).
- [28] Heather M. Russell and Julianna Tymoczko, *The transition matrix between the Specht and $\mathfrak{sl}_3$ web bases is unitriangular with respect to shadow containment*, Int. Math. Res. Not. IMRN **2022** (2022), no. 5, 3371–3416.
- [29] Heather M. Russell and Julianna S. Tymoczko, *The transition matrix between the Specht and web bases is unipotent with additional vanishing entries*, Int. Math. Res. Not. IMRN **2019** (2019), no. 5, 1479–1502.
- [30] Bruce E. Sagan, *The symmetric group. Representations, combinatorial algorithms, and symmetric functions*, second ed., Graduate Texts in Mathematics, vol. 203, Springer-Verlag, New York, 2001.
- [31] Bernd Sturmfels, *Algorithms in invariant theory*, second ed., Texts and Monographs in Symbolic Computation, SpringerWienNewYork, Vienna, 2008.
- [32] H. N. V. Temperley and E. H. Lieb, *Relations between the “percolation” and “colouring” problem and other graph-theoretical problems associated with regular planar lattices: some exact results for the “percolation” problem*, Proc. Roy. Soc. London Ser. A **322** (1971), no. 1549, 251–280.
- [33] Bruce W. Westbury, *Web bases for the general linear groups*, J. Algebraic Combin. **35** (2012), no. 1, 93–107.

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